

Coverage Report for controller

Table of Contents

- 1. [Analysis Information](#)
- 2. [Tests](#)
- 3. [Summary](#)
- 4. [Details](#)

Analysis Information

Coverage Data Information

Collected in version (R2023b)

Model Information

Model version 5.13
Author Jacopo Sini
Last saved Mon Jun 24 18:32:04 2024

Harness information

Harness model(s) IntegrationTest
Harness model owner HarnessTest

Simulation Optimization Options

Default parameter behavior tunable
Block reduction forced off
Conditional branch optimization on

Coverage Options

Analyzed model controller
Logic block short circuiting off
MCDC mode masking

Tests

Test	Started execution	Ended execution
Run 26	24-Jun-2024 18:41:37	24-Jun-2024 18:41:43

Summary

Model Hierarchy/Complexity	Run 26				Execution
	Decision	Condition	MCDC		
1. controller	37 20%	21%	16%		NA
2. . . . Subsystem	36 20%	21%	16%		NA
3. OnePedalController	30 6%	3%	0%		NA
4. SF: Subsystem/OnePedalController	29 6%	3%	0%		NA
5. SF: B	8 0%	NA	NA		NA
6. SafetyFunctions	6 100%	88%	75%		NA
7. SF: Subsystem/SafetyFunctions	5 100%	88%	75%		NA

Details

1. Model "controller"

Child Systems: [Subsystem](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	37
Decision	NA	20% (8/41) decision outcomes
Condition	NA	21% (8/38) condition outcomes
MCDC	NA	16% (3/19) conditions reversed the outcome
Execution	NA	NA

2. SubSystem block "[Subsystem](#)"

[Justify or Exclude](#)

Parent: [/controller](#)

Child Systems: [OnePedalController](#), [SaftyFunctions](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	0	36
Condition	NA	21% (8/38) condition outcomes
Decision	NA	20% (8/41) decision outcomes
MCDC	NA	16% (3/19) conditions reversed the outcome

3. SubSystem block "[OnePedalController](#)"

[Justify or Exclude](#)

Parent: [controller/Subsystem](#)

Child Systems: [Subsystem/OnePedalController](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	30
Condition	NA	3% (1/30) condition outcomes
Decision	NA	6% (2/35) decision outcomes
MCDC	NA	0% (0/15) conditions reversed the outcome

4. Chart "[Subsystem/OnePedalController](#)"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)

Child Systems: [B](#)

Uncovered Links: 

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	4	29
Condition	NA	3% (1/30) condition outcomes
Decision	20% (1/5) decision outcomes	6% (2/35) decision outcomes
MCDC	NA	0% (0/15) conditions reversed the outcome

Decisions analyzed

Substate executed	20%
State "B"	0/8000 
State "D"	0/8000 
State "N"	0/8000 
State "P"	8000/8000
State "R"	0/8000 

Transition "[AutomaticTransmissionSelectorState~Tran...](#)" from "[P](#)" to "[N](#)"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)

Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	2
Condition	25% (1/4) condition outcomes
Decision	50% (1/2) decision outcomes
MCDC	0% (0/2) conditions reversed the outcome

1 [AutomaticTrasmissionSelectorState~=TransmissionState.Park&BrakePedalPressed==1]

#1: |AutomaticTrasmissionSelectorState~=TransmissionState.Park&BrakePedalPressed...

Decisions analyzed

AutomaticTrasmissionSelectorState~=TransmissionState.Park&BrakePedalPressed==1	50%
false	8000/8000
true	0/8000

Conditions analyzed

Description	True	False
AutomaticTrasmissionSelectorState~=TransmissionState.Park	0	8000
BrakePedalPressed==1	0	0

MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition	True Out	False Out
AutomaticTrasmissionSelectorState~=TransmissionState.Park&BrakePedalPressed==1		
AutomaticTrasmissionSelectorState~=TransmissionState.Park	(TT)	Fx
BrakePedalPressed==1	(TT)	(TF)

Transition "|AutomaticTrasmissionSelectorState==Tran..." from "N" to "P"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)

Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	3
Condition	0% (0/6) condition outcomes
Decision	0% (0/2) decision outcomes
MCDC	0% (0/3) conditions reversed the outcome

1 [AutomaticTrasmissionSelectorState==TransmissionState.Park&BrakePedalPressed==1&abs(VehicleSpeed_kmh)<5]

#1: |AutomaticTrasmissionSelectorState==TransmissionState.Park&BrakePedalPressed...

Decisions analyzed

AutomaticTrasmissionSelectorState==TransmissionState.Park&BrakePedalPressed==1&abs(VehicleSpeed_kmh)<5	0%
false	--
true	--

Conditions analyzed

Description	True	False
AutomaticTrasmissionSelectorState==TransmissionState.Park	0	0
BrakePedalPressed==1	0	0

abs(VehicleSpeed_km_h)<5	0 	0
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MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition	True Out	False Out
AutomaticTransmissionSelectorState==TransmissionState.Park&BrakePedalPressed==1&abs(VehicleSpeed_km_h)<5		
AutomaticTransmissionSelectorState==TransmissionState.Park	(TTT)	(Fxx)
BrakePedalPressed==1	(TTT)	(TFx)
abs(VehicleSpeed_km_h)<5	(TTT)	(TTF)

Transition "[|AutomaticTransmissionSelectorState==Tran...](#)" from "[N](#)" to "[R](#)"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)

Uncovered Links:

Metric	Coverage
Cyclomatic Complexity	3
Condition	0% (0/6) condition outcomes
Decision	0% (0/2) decision outcomes
MCDC	0% (0/3) conditions reversed the outcome

[1](#) [[AutomaticTransmissionSelectorState==TransmissionState.Reverse&BrakePedalPressed==1&VehicleSpeed_km_h<5](#)]

[#1: |AutomaticTransmissionSelectorState==TransmissionState.Reverse&BrakePedalPres...](#)

Decisions analyzed

AutomaticTransmissionSelectorState==TransmissionState.Reverse&BrakePedalPressed==1&VehicleSpeed_km_h<5	0%
false	--
true	--

Conditions analyzed

Description	True	False
AutomaticTransmissionSelectorState==TransmissionState.Reverse	0 	0
BrakePedalPressed==1	0 	0
VehicleSpeed_km_h<5	0 	0

MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition	True Out	False Out
AutomaticTransmissionSelectorState==TransmissionState.Reverse&BrakePedalPressed==1&VehicleSpeed_km_h<5		
AutomaticTransmissionSelectorState==TransmissionState.Reverse	(TTT)	(Fxx)
BrakePedalPressed==1	(TTT)	(TFx)
VehicleSpeed_km_h<5	(TTT)	(TTF)

Transition "[|AutomaticTransmissionSelectorState~Tran...](#)" from "[R](#)" to "[N](#)"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)

Uncovered Links:

Metric	Coverage
Cyclomatic Complexity	1
Decision	0% (0/2) decision outcomes

1 [AutomaticTrasmissionSelectorState~=TransmissionState.Reverse]

#1: [AutomaticTrasmissionSelectorState~=TransmissionState.Reverse]

Decisions analyzed

AutomaticTrasmissionSelectorState~=TransmissionState.Reverse	0%
false	--
true	--

Transition "[AutomaticTrasmissionSelectorState==Tran..." from "N" to "D"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)

Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	2
Condition	0% (0/4) condition outcomes
Decision	0% (0/2) decision outcomes
MCDC	0% (0/2) conditions reversed the outcome

1 [AutomaticTrasmissionSelectorState==TransmissionState.Drive|[AutomaticTrasmissionSelectorState==TransmissionState.Brake&BrakePedalPressed==1&VehicleSpeed_

#1: [AutomaticTrasmissionSelectorState==TransmissionState.Drive|[AutomaticTrasmi...

Decisions analyzed

AutomaticTrasmissionSelectorState==TransmissionState.Drive [AutomaticTrasmissionSelectorState==TransmissionState.Brake&BrakePedalPressed==1&VehicleSpeed_
false
true

Conditions analyzed

Description	True	False
AutomaticTrasmissionSelectorState==TransmissionState.Drive	0	0
AutomaticTrasmissionSelectorState==TransmissionState.Brake&BrakePedalPressed==1&VehicleSpeed_kmh>-5	0	0

MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition
AutomaticTrasmissionSelectorState==TransmissionState.Drive [AutomaticTrasmissionSelectorState==TransmissionState.Brake&BrakePedalPressed==1&VehicleSpeed_
AutomaticTrasmissionSelectorState==TransmissionState.Drive
AutomaticTrasmissionSelectorState==TransmissionState.Brake&BrakePedalPressed==1&VehicleSpeed_kmh>-5

Transition "[AutomaticTrasmissionSelectorState==Tran..." from "D" to "N"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)

Uncovered Links: 

Metric	Coverage
--------	----------

Cyclomatic Complexity	3
Condition	0% (0/6) condition outcomes
Decision	0% (0/2) decision outcomes
MCDC	0% (0/3) conditions reversed the outcome

1 [AutomaticTrasmissionSelectorState==TransmissionState.Neutral || AutomaticTrasmissionSelectorState==TransmissionState.Reverse || AutomaticTrasmissionS

#1: [AutomaticTrasmissionSelectorState==TransmissionState.Neutral || AutomaticTr...

Decisions analyzed

AutomaticTrasmissionSelectorState==TransmissionState.Neutral AutomaticTrasmissionSelectorState==TransmissionState.Reverse AutomaticTrasmissionSelectorState==TransmissionState.Park	0%
false	-- 
true	-- 

Conditions analyzed

Description	True	False
AutomaticTrasmissionSelectorState==TransmissionState.Neutral	0 	0 
AutomaticTrasmissionSelectorState==TransmissionState.Reverse	0 	0 
AutomaticTrasmissionSelectorState==TransmissionState.Park	0 	0 

MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition
AutomaticTrasmissionSelectorState==TransmissionState.Neutral AutomaticTrasmissionSelectorState==TransmissionState.Reverse AutomaticTrasmissionSelectorState==TransmissionState.Park
AutomaticTrasmissionSelectorState==TransmissionState.Neutral
AutomaticTrasmissionSelectorState==TransmissionState.Reverse
AutomaticTrasmissionSelectorState==TransmissionState.Park

Transition "[ThrottlePedalPosition>(1/3)&AutomaticT..." from "D" to "B"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)
Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	2
Condition	0% (0/4) condition outcomes
Decision	0% (0/2) decision outcomes
MCDC	0% (0/2) conditions reversed the outcome

1 [ThrottlePedalPosition>(1/3)&AutomaticTrasmissionSelectorState==TransmissionState.Brake]

#1: [ThrottlePedalPosition>(1/3)&AutomaticTrasmissionSelectorState==Transmissio...

Decisions analyzed

ThrottlePedalPosition> (1/3)&AutomaticTrasmissionSelectorState==TransmissionState.Brake	0%
false	-- 
true	-- 

Conditions analyzed

--	--	--

Description	True	False
ThrottlePedalPosition>(1/3)	0 	0 
AutomaticTrasmissionSelectorState==TransmissionState.Brake	0 	0 

MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition	True Out	False Out
ThrottlePedalPosition>(1/3)&AutomaticTrasmissionSelectorState==TransmissionState.Brake		
ThrottlePedalPosition>(1/3) 	(TT)	(Fx)
AutomaticTrasmissionSelectorState==TransmissionState.Brake 	(TT)	(TF)

Transition "[|AutomaticTrasmissionSelectorState~TransmissionState.Brake](#)" from "[B](#)" to "[D](#)"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)

Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	1
Decision	0% (0/2) decision outcomes

[1](#) [[AutomaticTrasmissionSelectorState~TransmissionState.Brake](#)]

[#1: |AutomaticTrasmissionSelectorState~TransmissionState.Brake](#)

Decisions analyzed

AutomaticTrasmissionSelectorState~TransmissionState.Brake	0% 
false	-- 
true	-- 

5. State "[B](#)"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController](#)

Uncovered Links: 

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	4	8
Decision	0% (0/6) decision outcomes	0% (0/14) decision outcomes

Decisions analyzed

Substate executed	0% 
State "Accelerate"	-- 
State "Decelerate"	-- 
State "Stop"	-- 
Substate exited when parent exits	0% 
State "Accelerate"	-- 
State "Decelerate"	-- 
State "Stop"	-- 

Transition "[\[ThrottlePedalPosition>=1/3\]](#)" from "[Stop](#)" to "[Accelerate](#)"

[Justify or Exclude](#)
Parent: [controller/Subsystem/OnePedalController.B](#)
Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	1
Decision	0% (0/2) decision outcomes

[1](#) [\[ThrottlePedalPosition>=1/3\]](#)

[#1: \[ThrottlePedalPosition>=1/3\]](#)

Decisions analyzed	
ThrottlePedalPosition>=1/3	0%
false	-- 
true	-- 

Transition "[\[ThrottlePedalPosition>\(1/3\)\]](#)" from "[Decelerate](#)" to "[Accelerate](#)"

[Justify or Exclude](#)
Parent: [controller/Subsystem/OnePedalController.B](#)
Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	1
Decision	0% (0/2) decision outcomes

[1](#) [\[ThrottlePedalPosition>\(1/3\)\]](#)

[#1: \[ThrottlePedalPosition>\(1/3\)\]](#)

Decisions analyzed	
ThrottlePedalPosition>(1/3)	0%
false	-- 
true	-- 

Transition "[\[ThrottlePedalPosition<=\(1/3\)\]](#)" from "[Accelerate](#)" to "[Decelerate](#)"

[Justify or Exclude](#)
Parent: [controller/Subsystem/OnePedalController.B](#)
Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	1
Decision	0% (0/2) decision outcomes

[1](#) [\[ThrottlePedalPosition<=\(1/3\)\]](#)

[#1: \[ThrottlePedalPosition<=\(1/3\)\]](#)

Decisions analyzed	
ThrottlePedalPosition<=(1/3)	0%
false	-- 
true	-- 

Transition "[VehicleSpeed_kmh<=0]" from "Decelerate" to "Stop"

[Justify or Exclude](#)

Parent: [controller/Subsystem/OnePedalController.B](#)

Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	1
Decision	0% (0/2) decision outcomes

1 [VehicleSpeed_kmh<=0]

#1: [VehicleSpeed_kmh<=0]

Decisions analyzed

VehicleSpeed_kmh<=0	0%
false	-- 
true	-- 

6. SubSystem block "SaftyFunctions"

[Justify or Exclude](#)

Parent: [controller/Subsystem](#)

Child Systems: [Subsystem/SaftyFunctions](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	6
Condition	NA	88% (7/8) condition outcomes
Decision	NA	100% (6/6) decision outcomes
MCDC	NA	75% (3/4) conditions reversed the outcome

7. Chart "Subsystem/SaftyFunctions"

[Justify or Exclude](#)

Parent: [controller/Subsystem/SaftyFunctions](#)

Metric	Coverage (this object)	Coverage (inc. descendants)
Cyclomatic Complexity	1	5
Condition	NA	88% (7/8) condition outcomes
Decision	100% (2/2) decision outcomes	100% (6/6) decision outcomes
MCDC	NA	75% (3/4) conditions reversed the outcome

Decisions analyzed

Substate executed	100%
State "SafeStateOFF"	1499/8000
State "StateStateON"	6501/8000

Transition "[abs(e)<=0.02&&CANDisconnection==0]" from "StateStateON" to "SafeStateOFF"

[Justify or Exclude](#)

Parent: [controller/Subsystem/SaftyFunctions](#)

Metric	Coverage
Cyclomatic Complexity	2
Condition	100% (4/4) condition outcomes
Decision	100% (2/2) decision outcomes
MCDC	100% (2/2) conditions reversed the outcome

1 [abs(e)<=0.02&&CANdisconnection==0]

#1: [abs(e)<=0.02&&CANdisconnection==0]

Decisions analyzed

abs(e)<=0.02&&CANdisconnection==0	100%
false	6500/6501
true	1/6501

Conditions analyzed

Description	True	False
abs(e)<=0.02	1501	5000
CANdisconnection==0	1	1500

MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition	True Out	False Out
abs(e)<=0.02&&CANdisconnection==0		
abs(e)<=0.02	TT	Fx
CANdisconnection==0	TT	TF

Transition "[abs(e)>=0.02||CANdisconnection==1]" from "SafeStateOFF" to "StateStateON"

Justify or Exclude

Parent: [controller/Subsystem/SafetyFunctions](#)

Uncovered Links: 

Metric	Coverage
Cyclomatic Complexity	2
Condition	75% (3/4) condition outcomes
Decision	100% (2/2) decision outcomes
MCDC	50% (1/2) conditions reversed the outcome

1 [abs(e)>=0.02||CANdisconnection==1]

#1: [abs(e)>=0.02||CANdisconnection==1]

Decisions analyzed

abs(e)>=0.02 CANdisconnection==1	100%
false	1498/1499
true	1/1499

Conditions analyzed

Description	True	False
abs(e)>=0.02	1	1498
CANdisconnection==1	0 	1498

MC/DC analysis (combinations in parentheses did not occur)

Decision/Condition	True Out	False Out
abs(e)>=0.02 CANdisconnection==1		
abs(e)>=0.02	Tx	FF
CANdisconnection==1 	(FT)	FF